

Team QuadCore

Job Search AI Agent

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Batch: [Batch ID]

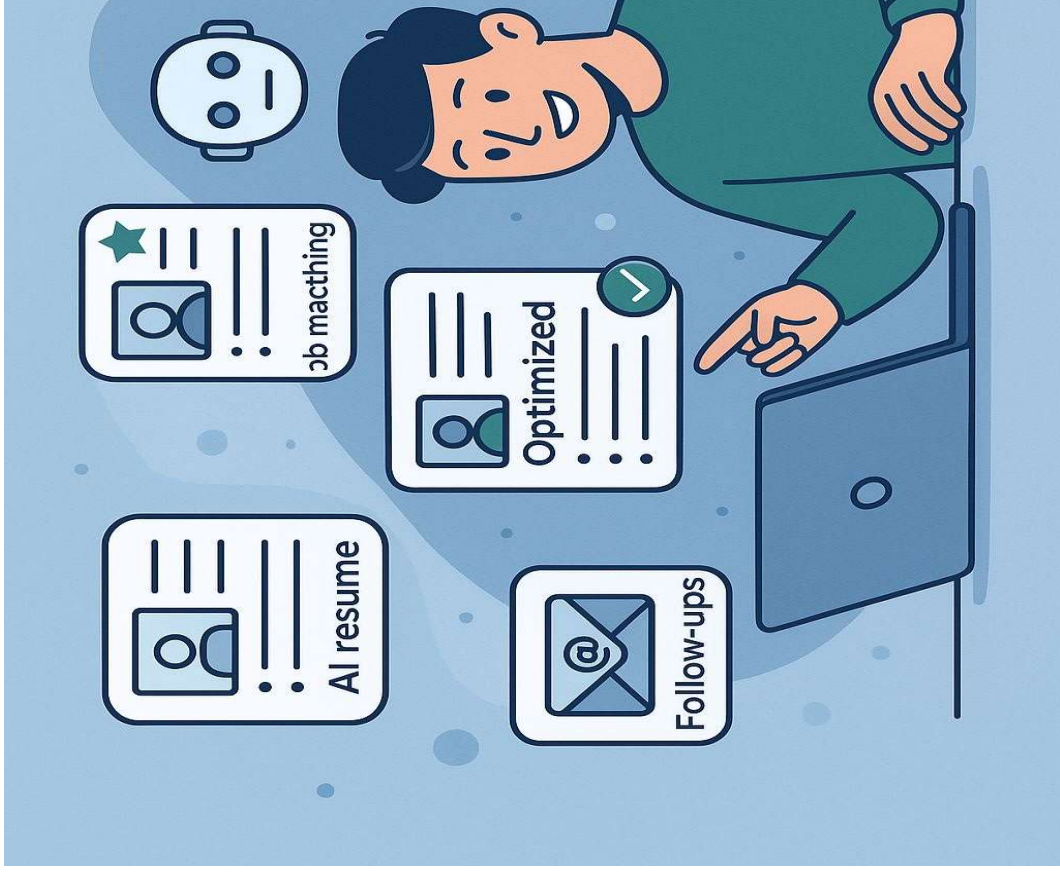
Duration: 2 Months / Tech: LangChain AI Agents

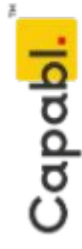
PROJECT OVERVIEW

Mission Statement

Our AI-Powered Job Search & Career Assistant automates job discovery and resume analysis using real job market data. It helps users find relevant opportunities through skill-based matching and AI-driven career guidance. The system is designed to be simple, reliable, and easy to use for job seekers.

- Core Focus: Career Development & Job Search Automation
- Main Langchain Module: LLM Wrapper: ChatOpenAI





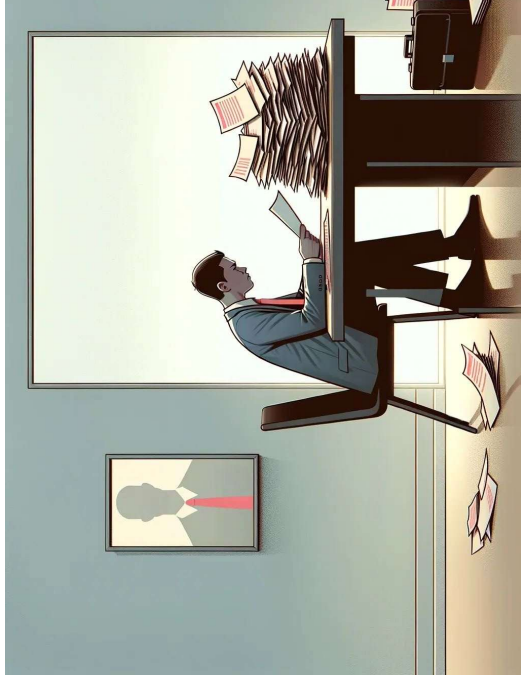
THE CHALLENGE

Identifying the Friction

Job seekers struggle with fragmented job platforms, manual resume screening, and difficulty identifying relevant opportunities.

Traditional scripts can fetch job listings but fail to understand resumes, match skills intelligently, or provide personalized career guidance.

This problem requires an AI agent to interpret unstructured resume data, adapt to user preferences, and assist users dynamically throughout the job search process.



Why Job Search Needs an AI-Powered

1 Fragmented Platforms

- Multiple job boards, overwhelming listings

2 Manual Filtering

- Time-consuming and error-prone

What's Needed:



Understand resumes



Adapt dynamically



Assist dynamically



TECHNOLOGY STACK



Framework

LangChain LLM Wrapper:
ChatOpenAI

Flask (Backend API)
Streamlit (Frontend UI)



LLM Engine

OpenAI GPT –
gpt-3.5-turbo



Vector DB

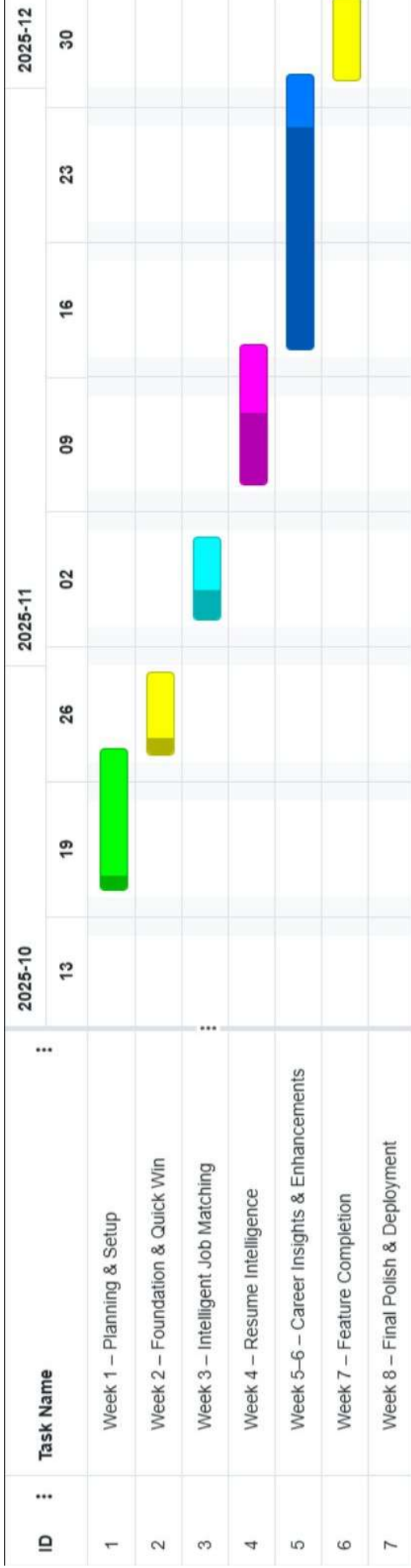
ChromaDB



Hosting

Local Deployment
Streamlit Cloud

TIMELINE



We followed a structured 8-week development timeline to build the AI-Powered Job Search & Career Assistant. In Week 1, we finalized the problem statement, designed the system architecture, set up the GitHub repository, and test prompts.

In Week 2, we built the foundation by implementing job search, integrating Streamlit with Flask, and deploying an initial version. In Week 3, we enhanced the system with resume-based skill extraction, intelligent job matching, and job relevance scoring. In Week 4, we focused on resume intelligence by adding ATS-friendly scoring, AI-based feedback, and resume improvement suggestions. During Weeks 5 and 6, we expanded the platform with career insights, improved ATS scoring, company information, salary history, and dashboard enhancements.

In Week 7, we completed all core features, optimized backend performance, and finalized a clean three-module interface. In Week 8, we performed full testing, UI polishing, bug fixing, deployment, and documentation.

Throughout the project, we followed an iterative development approach with continuous testing and improvements. This structured execution allowed us to deliver a complete, reliable, and deployable Track A career assistant.

SYSTEM ARCHITECTURE

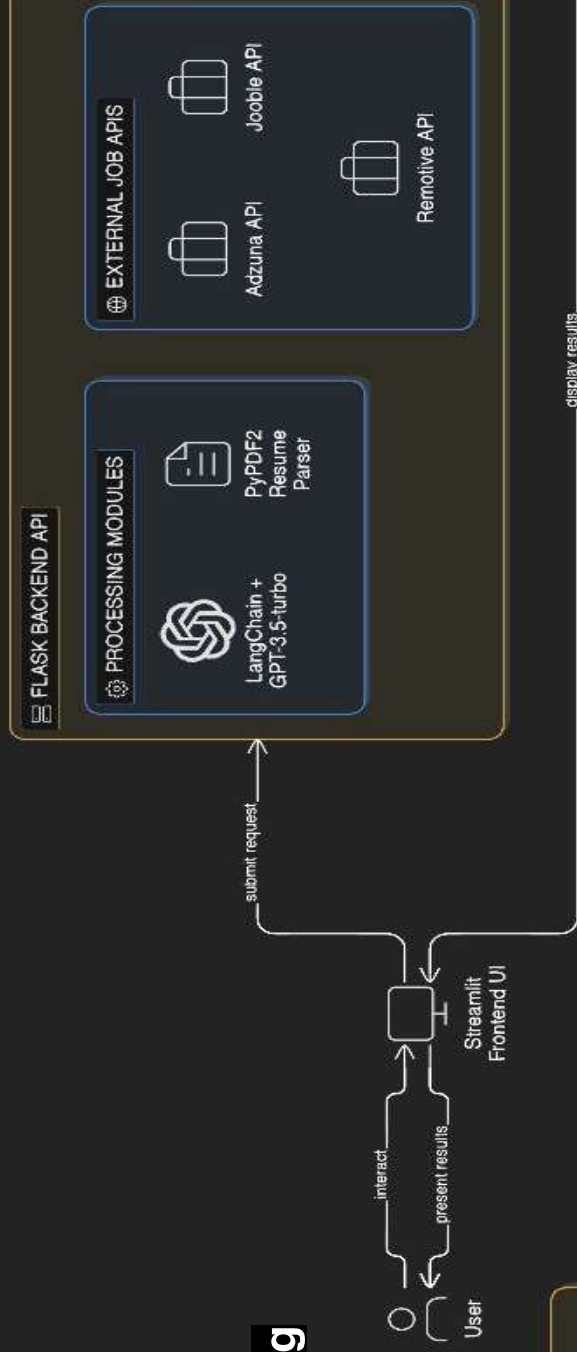
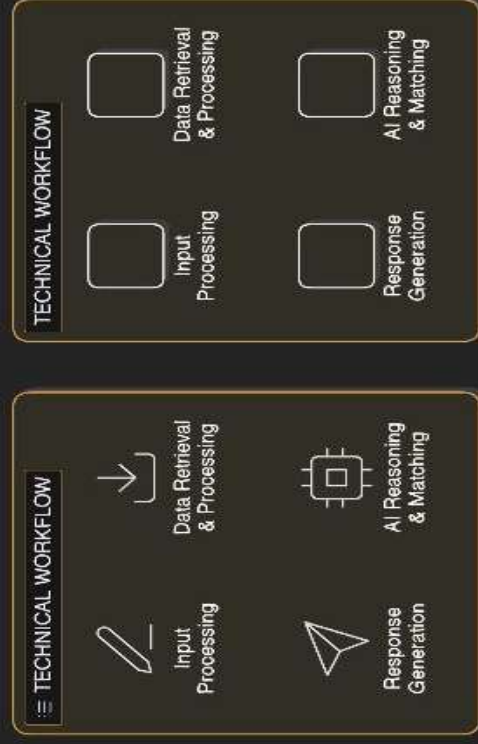
Technical Workflow

Step 1: Input Processing

Step 2: Retrieval & Tool Logic

Step 3: AI Reasoning & Matching

Step 4: Response Generation



Our system follows a **deterministic request-response** rather than a continuous agentic loop.

AI reasoning is applied at key stages for resume matching, and insight generation, ensuring exp reliable outputs.

NOVEL APPROACH



Unique

What makes your project stand out?

Resume-first job search: Jobs are discovered and ranked by resume, not just search keywords.

ATS score that actually matters: ATS scoring directly influences skill-gap analysis, not shown as a standalone metric.

Explainable job matching: Users can see *which* skills match recommended.

Single-action, multi-output pipeline: One input generates resume feedback, and company insights together.

Indian job market-aware logic: Filters and insights reflect like salary ranges, WFH roles, and company ratings.

DEMO VIDEO

Introduction

- Project: AI-Powered Job Search & Career Assistant
- Team Name: QuadCore
- Team of 4 Members-
Mallikarjunayya, Gururaj, Gangadhar & Akshay
- Introduction, Problem, Solution & UI C



ROADMAP & FUTURE SCOPE



Phase 1



Phase 2



Phase 3

Smarter Matching Logic

Feedback-Driven Ranking

UI/UX Enhancements





Conclusion

Ultimately, our project demonstrates that LangChain-powered assistants can reduce job search friction and decision latency by automating resume-driven matching and insight generation without relying on complex agentic systems.

We chose this problem because current job market systems are highly fragmented and inefficient, especially for early-career software engineers who rely heavily on poorly managed college placement processes.

By creating a unified, resume-first job discovery workflow, our system aims to provide a more transparent, accessible, and structured way for candidates to identify relevant opportunities beyond traditional placement channels.

THANK YOU FOR YOUR ATTENTION

Questions & Feedback are welcome.